

Task: GRZ - On Mushrooms!

Lab 3. Available memory: 512 MB. Deadline: 16.11.2025, 23:59:59

Description

Autumn has arrived, and Bajtazar can finally indulge in his favorite activity: picking mushrooms in the nearby forest[cite: 4].

The forest consists of trees forming an $n \times m$ grid. To avoid getting lost, Bajtazar always starts collecting mushrooms at tree $(1, 1)$ and finishes at tree (n, m) [cite: 5].

Standing at tree (i, j) , he can move in one step to tree $(i + 1, j)$, $(i - 1, j)$, or $(i, j + 1)$, assuming these trees exist[cite: 6]. Bajtazar never visits the same tree twice. We call a route with these properties a *mushroom picking*[cite: 7].

At most one mushroom grows under each tree. Bajtazar knows the forest perfectly, including the location of all mushrooms[cite: 8]. This allows him to hope that he can collect at least k mushrooms: in such a case, we say the mushroom picking is *successful*[cite: 9]. We assume that if Bajtazar passes a tree with a mushroom, he will definitely pick it[cite: 10].

Bajtazar wonders how many successful mushroom pickings exist. Since he dislikes large numbers, the last 9 digits of the result are sufficient[cite: 11].

Input

The first line of input contains four integers: n, m, k, g ($1 \leq n, m \leq 1000$, $0 \leq k \leq 10$, $0 \leq g \leq nm$), representing the dimensions of the forest, the minimum number of mushrooms k to collect, and the total number of mushrooms in the forest[cite: 13].

Each of the following g lines contains two integers a, b separated by a space ($1 \leq a \leq n$, $1 \leq b \leq m$), denoting a mushroom at the tree with coordinates (a, b) . All pairs of coordinates are distinct[cite: 14].

Output

Your program should print a single integer: the last 9 digits of the number of successful mushroom pickings. (Leading zeros should be omitted) [cite: 16].

Example

Input:

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2 3 2 3
1 2
2 1
2 2
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Output:

Explanation: The forest corresponding to the input data has 4 possible paths. All except the first one are successful[cite: 27, 28].